



Trimble® Earthworks Dozer

For SITECH Only

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Dozer hardware install checklists

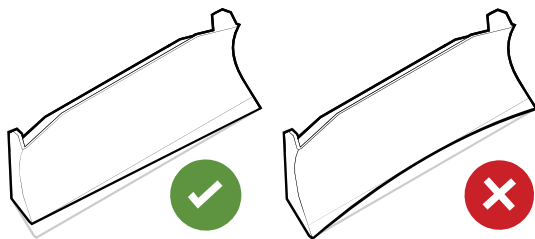
In this guide:

▶ Checklist—Pre-installation

1.1 Checklist—Pre-installation

Use the following checks to identify problems with a machine before an installation. Problems that are identified by this procedure must be corrected before commissioning is begun. Poor machine condition can affect the performance of the automatic blade guidance.

- ❑ Operate the machine and observe the performance of the hydraulic system, particularly the responsiveness of the hydraulic system to control movements. Any hydraulic system issues will remain after installation, and may be worsened if an Autos system is installed.
- ❑ Check the cutting edge:
 - Even vertical wear over the entire length of the blade. 0 ± 5 mm horizontal deviation.



- ❑ Check the track tension: refer to the machine manufacturer's operator manual for guidance.
- ❑ Check the hydraulic rams and joints: place the blade on flat ground. Activate the rams to raise the blade and ensure there is limited play in the joints.
- ❑ On C-frame machines, check for wear on the ball joint.
- ❑ On EH machines, for machine ECM firmware:
 - Contact the machine manufacturer to ensure that the ECM firmware is up to date.
- ❑ On EH machines, perform a:
 - Joystick calibration
 - Solenoid or cracking current calibration

NOTE –Front idler height can also affect Autos performance. This requires dealer support to assess and adjust.

Tools and other supplies

In this guide:

- ▶ Replacement serial number plates
- ▶ Extra extension cables
- ▶ Tools

2.1 Replacement serial number plates

If you need to install a system component over a serial number plate on the machine, you can order a replacement serial number plate from your dealer. You can then install the replacement serial number plate in another location on the machine. To maintain your machine's identification, you must obtain your dealer's approval to relocate the serial number plate before removing the existing serial number plate.

2.2 Extra extension cables

Always take a extra extension cables of varying lengths to an installation. The extension cables included in the kit may not be long enough for each machine's specific circumstances.

2.3 Tools

Make sure that you have the following tools and supplies available:

- Arc Welder / Welding Rods
- Volt/Ohm Meter
- Mechanical and Electrical hand Tools
- Clean Container for Hydraulic Oil
- Clean Rags

Dozer cab install guide

In this guide:

- ▶ Overview
- ▶ Component mounting locations
- ▶ Power cable routing
- ▶ EC520 control module install
- ▶ TD510/TD520 display install
- ▶ TD540 display install
- ▶ CI5xx CAN interface module install
- ▶ Valve module install
- ▶ Installing the AA510 audible alarm
- ▶ External Lightbar install (optional)
- ▶ Remote switch install

3.1 Overview

This guide describes how to install the components in and around the cab for the generic blade mount dozer system. For more detailed hardware and cabling installation instructions, refer to the machine-specific instructions that accompany the component kits.

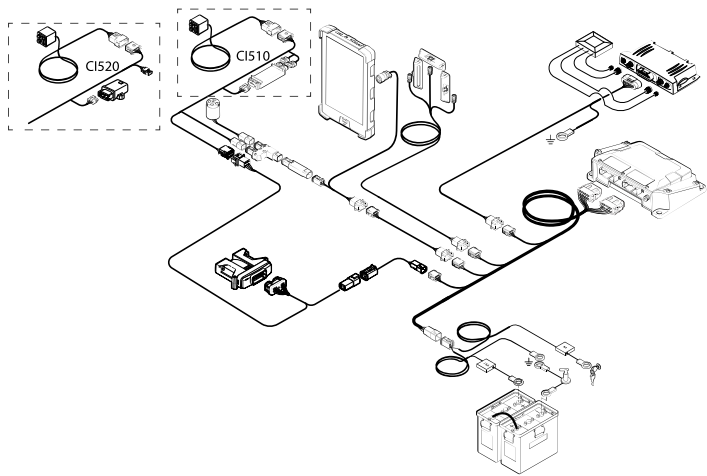
For Inactive Body IMU dozer cab mounted track mapping installation instructions, refer to the machine-specific instruction that accompany the component kits.

3.1.1 Cable routing best practice

When selecting a route for cables and harnessing consider the following:

- Exposure to hazards, including:
 - mechanical damage by the operator
 - environmental hazards
 - heat
 - corrosive material
- Moving parts
- Sharp edges

3.1.2 Cab components, cables and connections



Some users may replace the TD5x0 with a supported third party tablet.

3.1.3 Kit list

Use the following kits to install the dozer cab components:

Kit number	Description
160005-500	Base, Dozer, Generic Blade Mount, V3.0
160005-501	Display Add-on, Dozer, Generic Blade Mount, V3.0
160005-502	Valve Module Add-on, Dozer, Generic Blade Mount, V3.0
160041-10	Kit - SNM941 roof mounting
112057-00	Kit - SNM941 4-in-1 antenna with mounting bracket
Supported third party tablet (optional)	You may swap the TD5x0 for a third party tablet. Talk to your dealer about which tablets are supported.

3.1.4 Prerequisites

Check before starting:

- If the machine is a 12V machine, order a 12V relay, P/N 109333-12, instead of the 24V relay, P/N 109333-24, that is in the kit.
- Do not remove the backing film or the sorbothane layer on the EC520. The film protects the sorbothane layer.

- If you are using a supported third party tablet as the display, there are some important differences in the install procedure. Refer to [Chapter 7, Supported third party tablet installation](#) for more information.

3.1.5 Connecting to machine power

NOTE –If you connect the system's ground (negative) connection to the negative terminal of the battery, and attempt to start the machine with the master disconnect switch open (or if the master disconnect switch or battery ground cable is defective), then the starter current will flow from a grounded component on the chassis to the battery terminal and damage the system. Make sure that you connect the system's ground connection to the machine frame.

TIP – While you are connecting the power, place a "Machine under service, do not turn on" tag on the battery isolation switch.

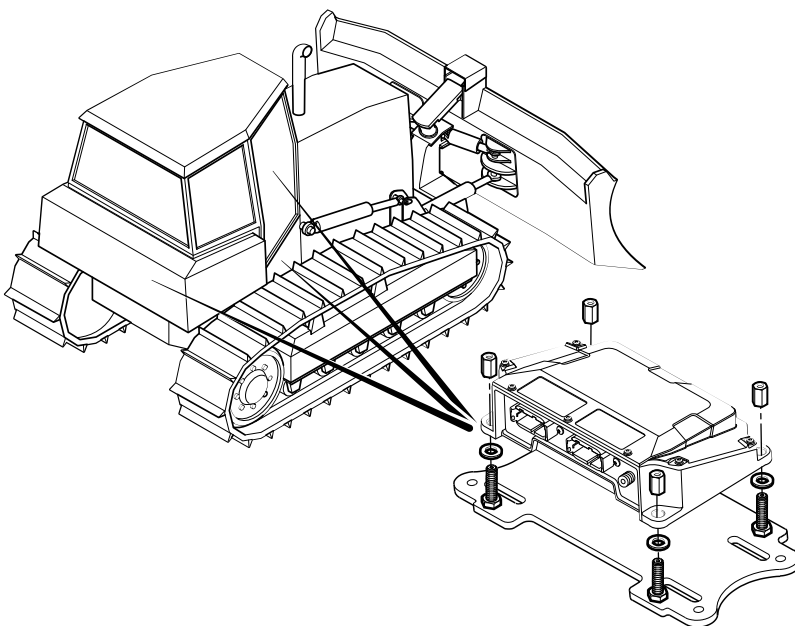
Constant power and key switched power connections are made on the main run relay for the machine, using the power cable.

The ground (Gnd) connection is made at the main chassis connection for the machine, located above the batteries.

3.2 Component mounting locations

This section shows possible locations of components in and around the cab. Read the install section for each component for hints and tips for installing that component, then choose the best location for the component on your machine.

3.2.1 EC520 control module



Requirements:

- Attach the EC520 bracket to the machine.
- Do not cut the cables. Use extension cables where needed.
- You cannot drill into, or weld on, any part of the machine's Roll Over Protection Structure (ROPS) as this could weaken its structural integrity.

NOTE –Many cabs are now an integral part of the machine's ROPS, and are certified as such. Drilling into, or welding on, these cabs may invalidate their ROPS certification.

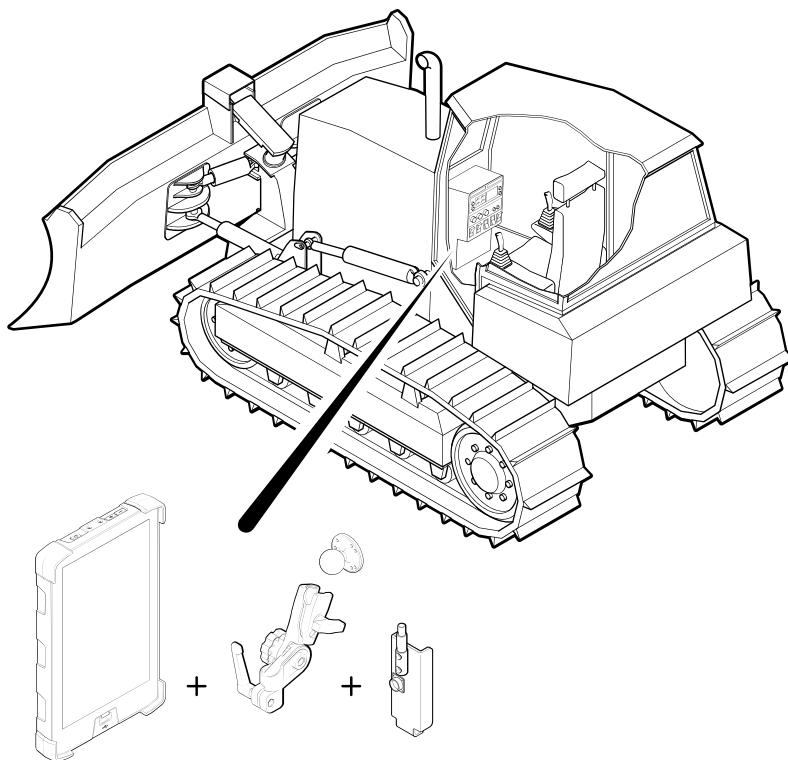
To avoid damage, consider:

- A location and orientation that prevents water collecting in the EC520 connectors.
- A location or surface that does not transfer heat to the EC520 or the harness.
- Space to route and bend cables from the EC520 into the cab, ensuring that you only bend cables within the tolerances specified in the instructions for the machine-specific component kits.

Suggested locations for the EC520:

- Directly under the cab, if such a surface exists. This may be above a plate protecting the under side of the cab.
- In the engine compartment.
- Behind the cab.

3.2.2 TD5x0 display



To select a route for the display cable consider the following:

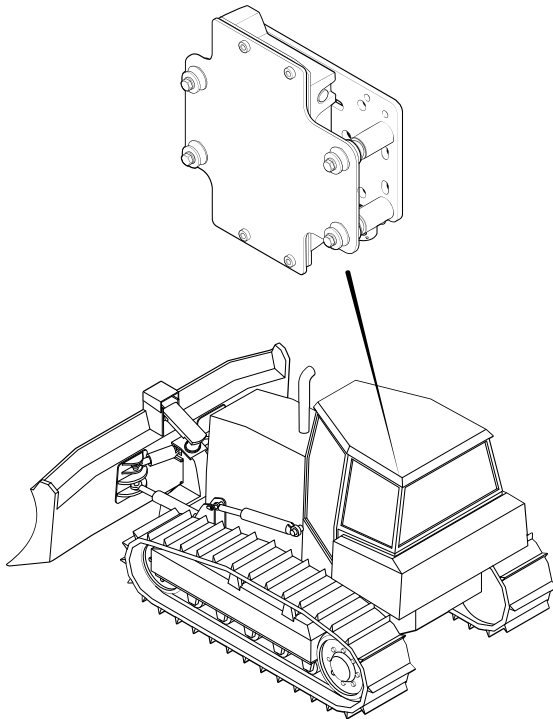
- Exposure to hazards, including:
 - mechanical damage by the operator
 - environmental hazards
 - heat
 - corrosive material
 - weather
- Existing grommated entry points. When taking the cable outside the cab, use grommated holes to minimize dust and moisture ingress to the cab.
- Maintaining free space between structures holding the display, or the display itself. For example, maintain free space between the display and windows that open and close, controls, and the view of the work area.
- Existing grommated entry points. When taking the cable outside the cab, use grommated holes to minimize the amount of dust and moisture in the cab.
- Maintaining free space between structures holding the display, or the display itself. Maintain free space between the system and windows that open and close, controls, and the view of the work area.
- Access to the remote switch cluster, joystick, and existing machine harness
- Access to the AA510

- Moving parts
- Sharp edges

Try various positions:

- Attach the Ram ratchet ball-clamp to the ball and the display and trial various positions.

3.2.3 SNM94x Connected Site Gateway



A 2D system does not need an SNM94x. A 3D system needs either an SNM94x or an SNRx21 or SNRx3x radio to receive GNSS corrections.

NOTE –Do not weld any components at this stage. Carefully mark the locations for the weld-mounted components, and then route the cables to make sure they will reach the components when they are installed.

NOTE –Follow the connection diagram to check the cable reaches the proposed location of the bracket. Do not cut the cable. Use extension cables where needed.

The primary consideration is for operator and technician safety:

- Rotate the SNM94x Connected Site Gateway so that the connector is along the most convenient side and you can access the SIM card. If you rotate the bracket, move the stowage blank appropriately.

For optimum performance:

- Connect any antennae above the roof line with an uninterrupted view.
- The lengths of the cable between antennae and the SNM94x Connected Site Gateway are fixed. Do not use extension cables as extension cables reduce performance. Do not shorten the antenna cables as cutting cables violates some regulations.
- To prevent signal interference the antennae must be at least 200 mm away from any other transmitting antenna, 100 mm away from any receiving antenna, and higher than surrounding metal objects.
- If using an SNR radio, the antenna must be 100 mm away from either of the SNM94x antennae.

To avoid damage:

- To reduce damage to the SNM94x from shock and vibration, position the bracket on a solid part of the cab.

Attach SNM94x Connected Site Gateway to bracket

For specific installation instructions, refer to the instructions provided with the radio bracket kit.

Validate

To check you have installed the SNM94x Connected Site Gateway and Wi-Fi antenna correctly, turn on the system.

- After a couple of minutes the SNM94x Connected Site Gateway will appear in the list of devices in the Web Interface.
- Check that the Web Interface's Monitor > Onboard Devices screen shows that the connection to the gateway is good.

3.2.4 Brackets for Wi-Fi antenna and cellular + GNSS antenna

The SNM941 uses a single integrated Wi-Fi antenna and cellular network antenna + L1 GNSS patch antenna.

3.2.5 Optional SNR radio

If using an SNM94x, you can mount the SNR radio on top of the gateway module.

If attaching the SNR radio to any other part of the machine, make sure the part is well braced as the extra weight of the radio and bracket can bend some machine parts.

Consider the following:

- Check that the radio cable reaches from the EC520 long platform harness to the radio bracket on the cab.
- The location of the SNR rover radio connector and the route for the SNR radio cable.

3.2.6 Valve module and CI5xx CAN interface module

The valve module and the CI5xx CAN interface module can be mounted inside the cab.

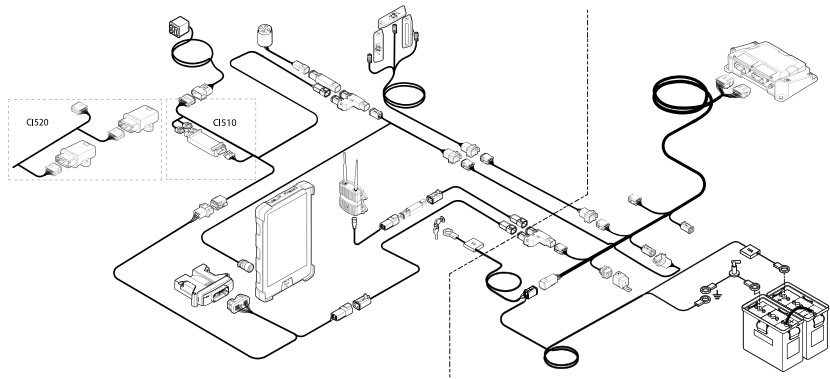
3.2.7 Installing SNM941 antenna

Install the SNM941 antenna to send and receive data to the office including, firmware updates, diagnostic files, position information and design files.

Refer to the *SNM941 Connected Site Gateway Installation Guide* for a detailed description of the modem installation.

3.3 Power cable routing

NOTE –Turn off the master disconnect switch before connecting to the power source on the machine.



3.3.1 Power connections, components, and cables

Part number	Items	Description
150603-02	1	EC520 long platform harness
150411-045	1	Power cable
109333-12	1	Relay* (12V)
109333-24	1	Relay* (24V)

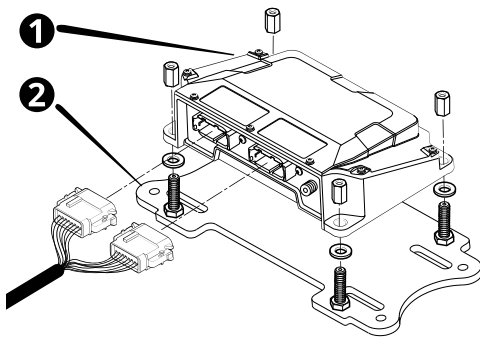
* The relay in the kit is for a 24V machine. If the machine is a 12V machine, order a 12V relay.

To connect:

- 1. Route the power cable, between the EC520 long platform harness and the battery.
- 2. Connect the relay, as shown in the connection diagram.

3.4 EC520 control module install

This section describes how to install the EC520 control module.



Item	Part number	Description
❶	100435-10	EC520
❷	201910-10	Bracket

3.4.1 Connections, components, and cables

Part number	Items	Description
150603-02	1	EC520 long platform harness
150411-045	1	Power cable

3.4.2 Install the EC520 control module

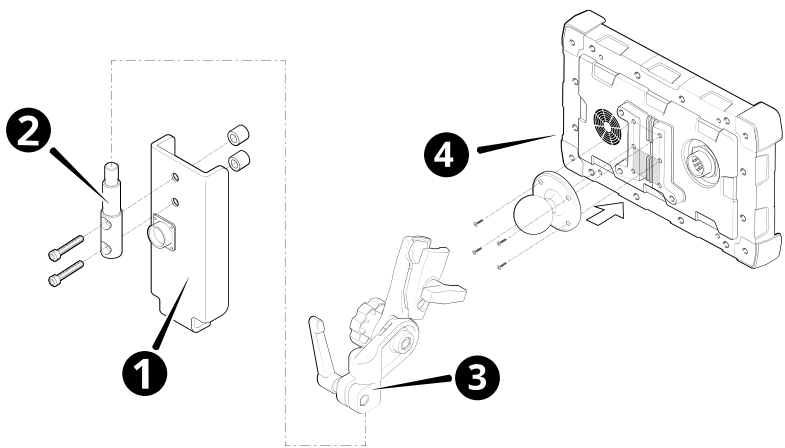
1. Decide on an installation location on the machine.
2. Place the bracket on the installation surface.
3. Using the bracket as a template, drill holes at the outside edges of the wide slots.
You cannot drill into, or weld on, any part of the machine's Roll Over Protection Structure (ROPS) as this could weaken its structural integrity.

NOTE –Many cabs are now an integral part of the machine's ROPS, and are certified as such. Drilling into, or welding on, these cabs may invalidate their ROPS certification.

4. Clamp the bracket to the machine.
5. Attach the EC520 to the bracket using the coupling nuts.
6. Connect the EC520 to the cabling as shown in the connection diagram. See [3.1.2 Cab components, cables and connections](#).

3.5 TD510/TD520 display install

This section describes how to install the TD510/TD520 display with the Ram adjustable ball-clamp.



Item	Part number	Description
❶	90115	Display mounting bracket
❷	0794-5720	Ram post
❸	58055-00	Ram mount
❹		TD510/TD520 display

3.5.1 Connections, components, and cables

Part number	Items	Description
150701-030	1	Display harness, 3 m

3.5.2 Install the TD510/TD520 display

- 1. Attach the Ram ratchet ball-clamp to the display.
- 2. Attach the bracket to the machine at the desired location.
- 3. Attach the Ram post to the bracket.
- 4. Attach the display with Ram mount to the ram post.
- 5. Attach the display cable and any extension cables required to reach the EC520 long platform harness as shown in the connection diagram. See [3.1.2 Cab components, cables and connections](#).

3.5.3 Validate

- 1. Connect the display cable to the TD5x0 display.
- 2. Turn on the ignition. The display starts automatically.
- 3. Tap the Earthworks icon. Earthworks opens.
- 4. If the log in screen appears, then the display is connected to the EC520.

Some of the problems that you might see include:

Problem	Description	Resolution
The display does not start.	There is no power to the display.	Check connectors between the display and the EC520. Use the electrical schematic to check the pin numbers from the battery.
Using my own display, and not the TD5x0, I cannot connect to the EC520.	The first time you run Earthworks, there is no Wi-Fi connection with the EC520.	Connect a laptop, or similar, to the EC520 long platform harness "Display, Eth1 + CAN2" connector. On the laptop, open the Web Interface, log in, and set up the Wi-Fi access point on the EC520.
On startup, I received a warning to upgrade the display operating system.	An important operating system update for the display is available.	Contact your system dealer to schedule the update.

3.6 TD540 display install

This section describes how to install the TD540 display. There are 2 installation options:

- With the magnetic display mount (P/N 110262-00) and Ram ratchet ball-clamp, for cab window installations.
- With the post mount (P/N 597-1269) and Ram ratchet ball-clamp, for mounting on a vertical pole, such as a grab bar.

WARNING –If you have a pacemaker and handle the magnetic display mount, the magnet can affect the function of the pacemaker and cause physical harm. Remain at least 30 cm (1 foot) from the magnetic display mount if you have a pacemaker.

WARNING –The magnets on the magnetic display mount are strong. If parts of your hands get between two magnets when you move them, the magnets may suddenly move and pinch or clamp the skin. When the magnets are close together, position your hands so they are not between the two magnets.

NOTE –It is easier to perform this task with two people.

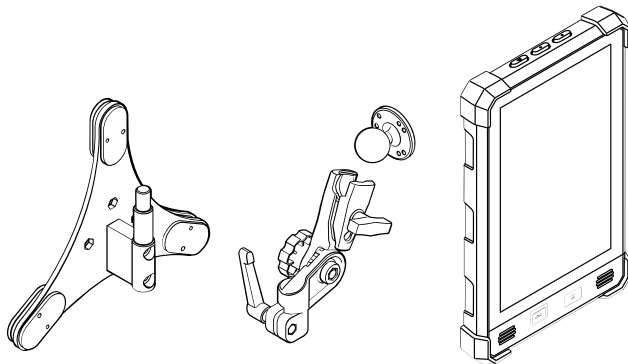
To select a route for the display cable, try various positions while considering:

- Exposure to hazards, including:
 - mechanical damage by the operator
 - environmental hazards

- heat
- corrosive material
- weather

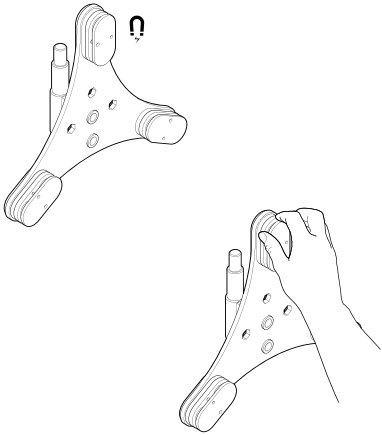
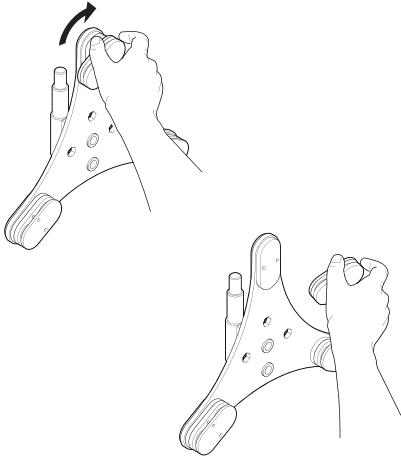
- Existing grommeted entry points. When taking the cable outside the cab, use grommeted holes to minimize dust and moisture ingress to the cab.
- Maintaining free space between structures holding the display, or the display itself. For example, maintain free space between the display and windows that open and close, controls, and the view of the work area.
- Access to the AA510 audible alarm
- Avoid moving parts
- Avoid sharp edges
- Operator's preference for location and orientation

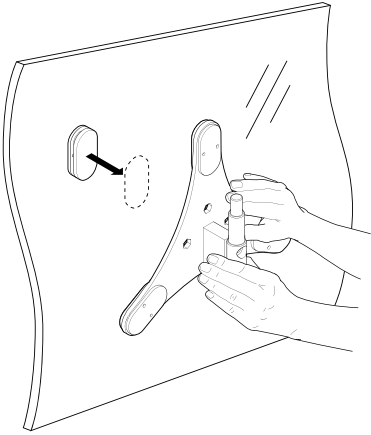
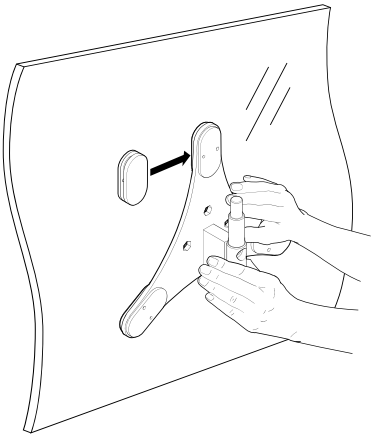
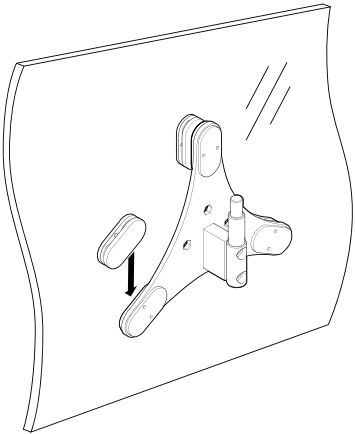
3.6.1 Option 1: Magnetic mount and display



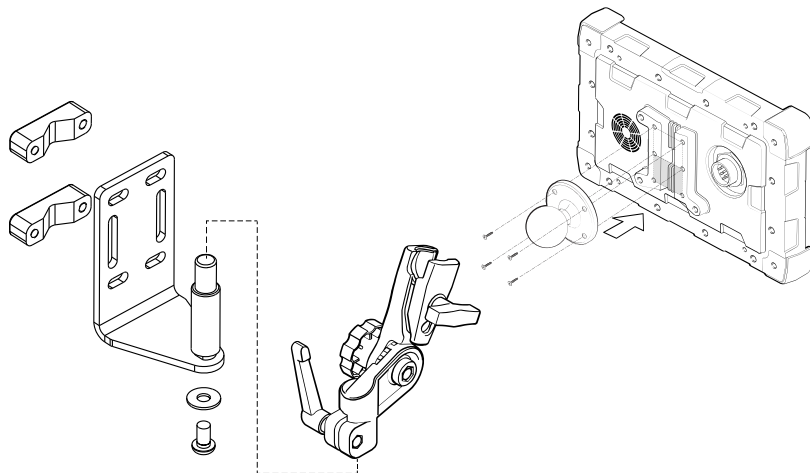
3.6.2 Attach magnetic mount and display

Steps	Description
The diagram shows a display unit mounted on a magnetic mount. A dashed line represents the 0° position, and a curved arrow indicates the display can be tilted up to +1-20°. An eye icon is shown looking at the display from the operator's perspective.	Decide on the location of the display.

Steps	Description
	Hold the magnetic display mount and carefully twist the magnet and tri-arm in opposite directions.
	When the force between the magnets weakens, pull the tri-arm and magnet apart. Remove the padding and keep the magnets away from each other.
Practice connecting and disconnecting.	Using two people and the padding, practice placing the tri-arm and magnet together. Keep the magnets close but not opposite each other with the padding in between, then rotate and slide the magnet until the magnets connect.

Steps	Description
	On the machine, place the tri-arm magnetic assembly against the inside cab window in the required position. Check the post is aligned correctly and that the cables, Ram ratchet ball-clamp, and display are in the correct position. Remove the Ram ratchet ball-clamp and display before continuing. Hold the interior tri-arm firmly in the required position and make sure all three magnets touch the glass.
	On the outside of the glass, attach the first magnet. • Once the magnets connect they are very hard to disconnect. • Keep the magnet surfaces away from dirt or other metals. • Start with the outside magnet away from the inside magnet, then slide toward each other.
	Repeat for all three magnets. Attach the Ram ratchet ball-clamp and then attach the display.

3.6.3 Option 2: Post mount and display

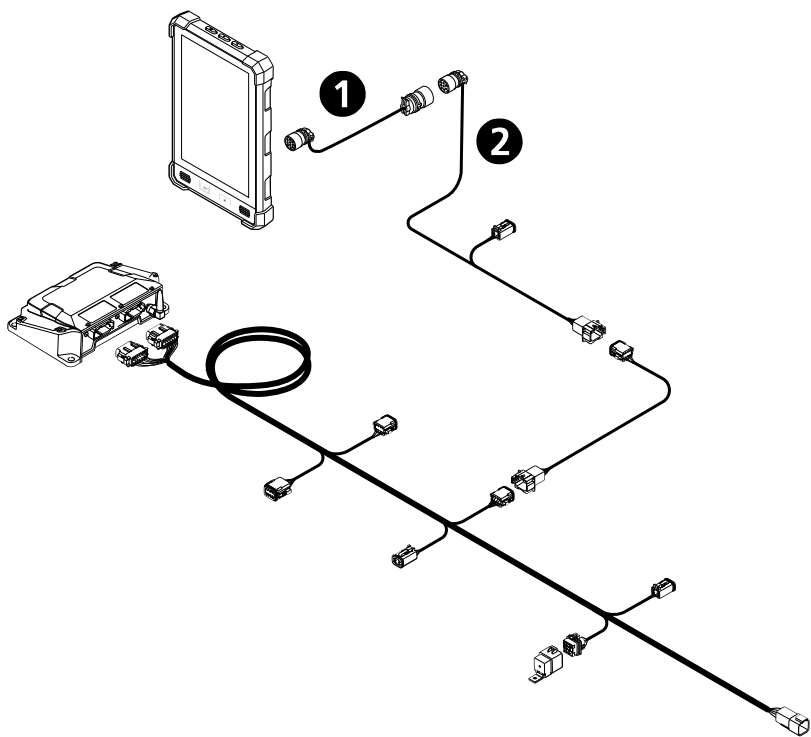


Attach the post mount

1. Connect the ball mount to the TD540.
2. Select a vertical post to mount the display to. The location should position the display within easy reach of the operator.
3. Use the appropriate M8 bolts and the V-blocks to attach the bracket to the post.
4. Fasten the RAM mount to the pole bracket.
5. Insert the ball mount into the RAM mount socket and fasten.

3.6.4 Connect the cables

1. Attach the TD540 adapter cable (P/N 150878-002, ❶) to the TD540 display. The adapter converts the display connector from 8-pin to 10-pin to match the display cable.
2. Attach the other end of the adapter cable to the TD5x0 display cable (P/N 150701-xxx, ❷).
3. Attach the display cable and any extension cables required to reach the EC520 long platform harness (P/N 150603-02).



3.6.5 Validate

1. Turn on the ignition. The display starts automatically.
2. Tap the Earthworks icon. Earthworks opens.
3. If the log in screen appears, then the display is connected to the EC520.

Some of the problems that you might see include:

Problem	Description	Resolution
The display does not start.	There is no power to the display.	Check connectors between the display and the EC520. Use the electrical schematic to check the pin numbers from the battery.
On startup, I received a warning to upgrade the display operating system.	An important operating system update for the display is available.	Contact your system dealer to schedule the update.

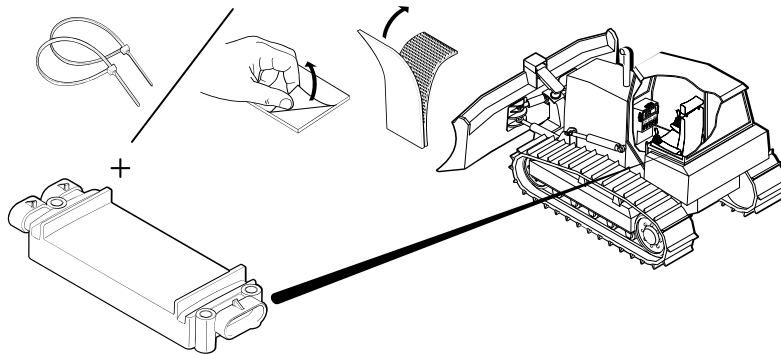
3.7 CI5xx CAN interface module install

The CI5xx is a Controller Area Network (CAN) adapter that converts the remote switch increment and decrement, and sideshift inputs to CAN.

Both the CI510 and the CI520 CAN interfaces are supported. The CI510 supports up to 8 switches. The CI520 supports up to 4 switches. Use two CI520s to connect 5-8 switches.

- [Attach CI510](#)
- [Attach CI520](#)

3.7.1 Attach CI510



3.7.2 CI510 connections, components, and cables

Part number	Items	Description
150906-01	1	Valve module harness
86930-05	1	Valve cable
09231-10	1	CI510 CAN Interface Module
150754-01		Switch harness

3.7.3 Install the CI510 CAN interface module

Have the following parts available:

- Dual locking strip
- Cable ties

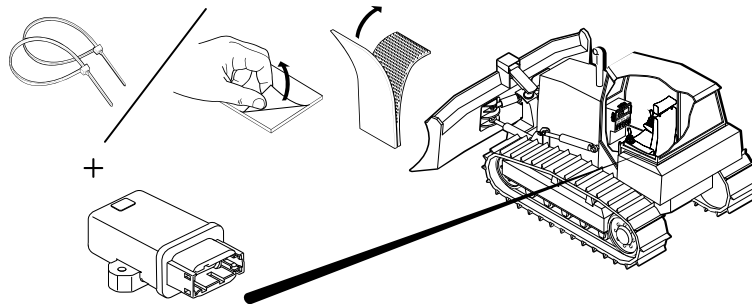
To install:

1. Connect the CI510 to the cabling as shown in the connection diagram. See [3.1.2 Cab components, cables and connections](#).
2. Place the CI510 inside, or under, the cab. It is often behind the seat.
3. Either use zip ties or use the Velcro supplied in the kit to secure the CI510, if required.

3.7.4 Validate

After connecting the display and running Earthworks, select System Status > Remote Switches. The remote switch status should be Active and the model should be CI510.

3.7.5 Attach CI520



3.7.6 CI520 connections, components, and cables

Part number	Items	Description
150906-01	1	Valve module harness
86930-05	1	Valve cable
132739-10	1	CI520 CAN Interface Module
150820-01 Rev B		Switch harness

3.7.7 Install the CI520 CAN interface module

Have the following parts available:

- Dual locking strip
- Cable ties

To install:

1. Connect the CI520 to the cabling as shown in the connection diagram. See [3.1.2 Cab components, cables and connections](#).

NOTE –The CI520 has only four switch inputs.

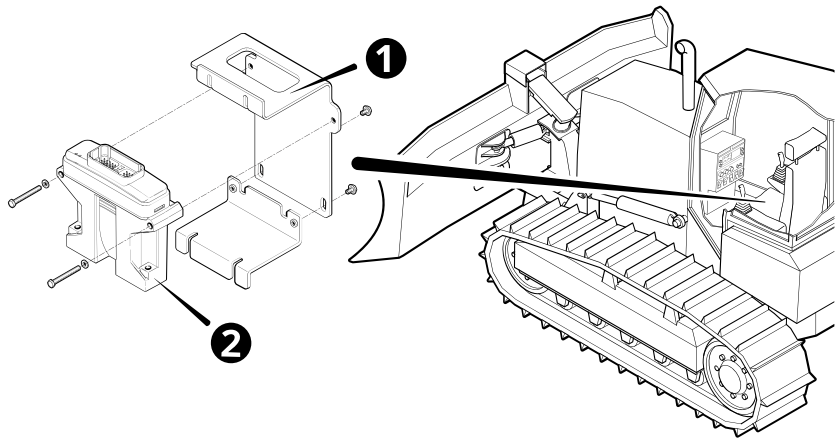
2. Place the CI520 inside, or under, the cab. It is often behind the seat.
3. Either use zip ties or use the Velcro supplied in the kit to secure the CI520, if required.

3.7.8 Validate

After connecting the display and running Earthworks, select System Status > Remote Switches. The remote switch status should be Active and the model should be CI520.

3.8 Valve module install

The valve module provides precise control of the blade's lift and tilt functions by sending output signals to the machine's electro-hydraulic valves or the hydraulic kit installed on the machine as part of this system.



Item	Part number	Description
①	196540-10	Module mounting bracket
②	100441-00	VM510

3.8.1 Connections, components, and cables

Part number	Items	Description
150906-01	1	Valve module harness
86930-05	1	Valve cable
150858-01	2	Splitter, 4 Circuit
150473-020	1	Cable extension

3.8.2 Install the valve module

1. Attach the valve module to the module mounting bracket.
2. Connect the cabling to and from the valve module according to the connection diagram for generic dozers, available on [Partners](#).

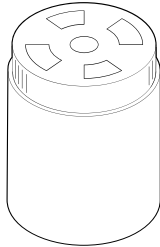
3.8.3 Validate

After connecting the display and running Earthworks, use the Web Interface's Configure > Valve Test screen to test that the valves are driving the lift and tilt rams.

3.9 Installing the AA510 audible alarm

The AA510 alerts the operator to whether they are on grade, above grade or below grade. The AA510 attaches to the system via a CAN bus.

3.9.1 Attach AA510



- Rubberized p-clamp P/N 97587-45
- Cable ties

To install:

1. Decide where to locate the AA510.
2. Connect the AA510 to the EC520 long platform harness, as shown in the connection diagram.
3. Use P-clamp or cable ties to attach the AA510 to a fixed surface or component.

3.9.2 Validate

After connecting the display and running Earthworks, test the AA510 using the System Settings > Sound option in Earthworks.

Some of the problems that you might see include:

Problem	Description	Resolution
AA510 too quiet or too loud.	The AA510 might seem loud in one machine and not loud enough in another machine because of where you locate the AA510 and the machine noise.	In the Operator App, open Sound Settings and edit the volume of the AA510 and the type of event that sets off the AA510.

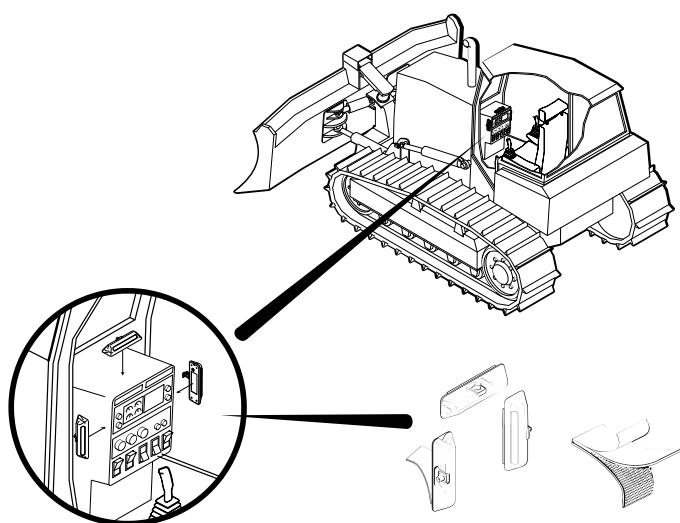
3.10 External Lightbar install (optional)

The optional external lightbars give extra guidance about being on grade, below grade, or above grade.

3.10.1 Connections, components, and cables

Part number	Items	Description
74706-01	3	LB400 lightbars in lightbar kit
54438-xx	1	Lightbar cable
54990	2	Industrial strength hook and loop tape (Velcro), in the Cab/Platform kit
150418-xx	1	Extension cable

3.10.2 Locate and attach lightbars



Each lightbar is 174 mm (6.85 in) long, 53 mm (2.1 in) wide, and has a height of 32 mm (1.3 in).

To install:

1. Route the lightbar cable and connect to the EC520 long platform harness, as shown in the connection diagram. See [3.1.2 Cab components, cables and connections](#).
2. Place the lightbars inside the machine cab using Velcro. Ensure that the LEDs on the lightbars are visible to the operator.

3.10.3 Validate

Do the following:

1. In the Web Interface, make sure you complete the entire Configure section, starting with the Install Assistant.
2. In Configure > External Lightbars, set the left, right, and center lightbars and select whether the lightbar mounting orientation is inverted.

3. In the Operator App, go to System Settings > Lightbars, use the slider to adjust the brightness of the lightbars.
4. Observe the lightbars during start up. Lightbars flash in a sequence when they start up.

The LEDs flash in the following sequence on start up:

LEDs on this lightbar	Flash
Left (blade tip cut/fill)	Bottom to top
Center (offline left/right)	Left to right
Right (blade tip cut/fill)	Top to bottom
All	Twice, simultaneously

Some of the problems that you might see include:

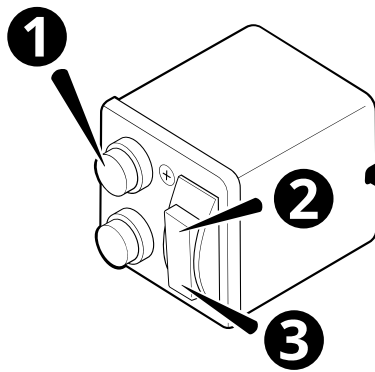
Problem	Description	Resolution
Lightbars do not illuminate	No power, or lightbar brightness is low.	Check that the display is on (to show power is on) and check the lightbar brightness is not set too low in the Operator App. Check the system harness between the EC520 and the lightbars.
Lightbars flash in the wrong sequence.	See sequences above.	The lightbars have been configured incorrectly. Open the Web Interface and check the configuration.

NOTE –For information about how lightbars work and how to operate them read the Operator's Guide, On-screen Lightbars, available online or in the Earthworks Operator App.

3.11 Remote switch install

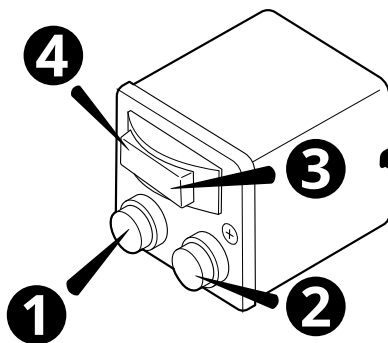
The dozer system can have up to two remote switches:

- Vertical offset remote switch:
 - activating the automatic controls
 - incrementing or decrementing the elevation offset or target slope



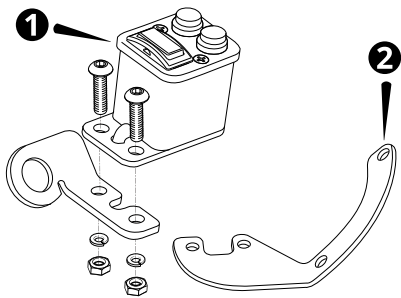
- | | |
|--------------------------------------|--------------------------------------|
| ❶ Auto switch. | ❷ Increment elevation offset switch. |
| <hr/> | |
| ❸ Decrement elevation offset switch. | |

- Horizontal offset remote switch:
 - incrementing or decrementing the horizontal offset
 - incrementing or decrementing the Pass Shift count



- | | |
|---|---|
| ❶ Decrement Pass Shift count switch. | ❷ Increment Pass Shift count switch. |
| <hr/> | |
| ❸ Increment Horizontal Distance D switch. | ❹ Decrement Horizontal Distance D switch. |

3.11.1 Remote switch assembly



Item	Part number	Description
1	150420-025	Remote switch assembly
2	0794-5920	Joystick handle mounting tab

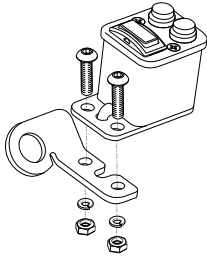
3.11.2 Install the switch harness

Select a route for the cable by considering:

- Exposure to hazards, including:
 - mechanical damage by the operator
 - environmental hazards
 - heat
 - corrosive material
- Access to the CI5xx cable
- Moving parts
- Sharp edges

3.11.3 Install and connect the remote switches

Illustration	Description
A line drawing showing a joystick handle with a remote switch assembly mounted vertically. The switch is connected to a cable. The joystick handle is shown in a cross-section view, and the switch is shown in a perspective view. The text 'Vertical offset remote switch install' is written next to the diagram.	Vertical offset remote switch install

Illustration	Description
	Horizontal offset remote switch install

To install:

1. Attach the supplied mounting tab to the base of the remote switch.

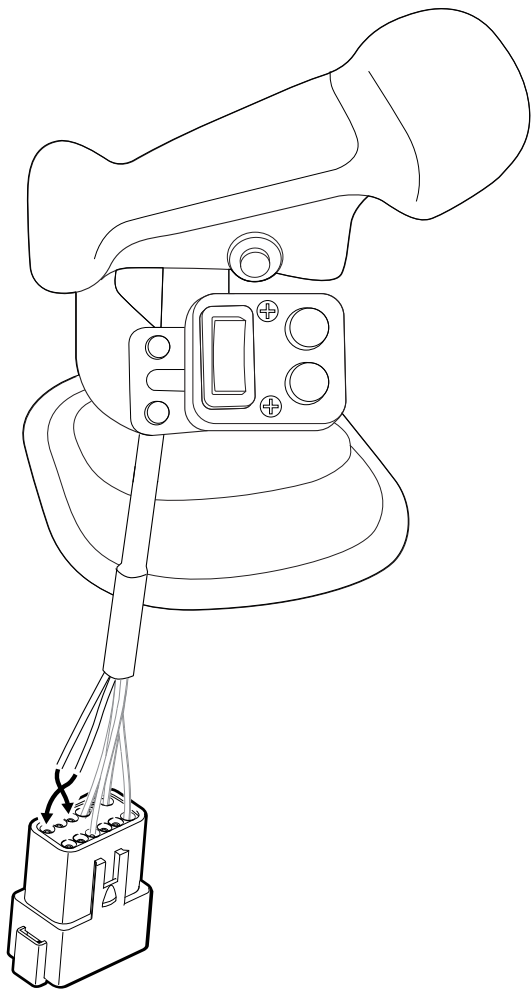
NOTE –If your joystick cannot be detached from the control assembly, replace the mounting tab in the kit with the joystick handle mounting tab.

2. Attach the mounting tab to the joystick.
3. Route and secure the remote control switch and connect the cable to the cab harness.

NOTE –If it is not practical to mount the remote switch on the blade control lever, you can make a custom mount.

Swap rocker

If you install the switch in an upside down orientation, you will need to reverse the direction of the increment/decrement rocker. To change the default rocker wiring so that it decrements when pressed down, swap the yellow (pin 10) and blue (pin 12) wires on the 12-pin deutsch connector.



3.11.4 Validate

Some of the problems that you might see include:

Problem	Description	Resolution
Remote switch does not work.	The switch is not connected.	Check the switch connects system by opening the Web Interface and checking Monitor > Onboard Devices. Check the connections.

Dozer main harness install guide

In this guide:

- ▶ Overview of the install
- ▶ Connections, components, and cables
- ▶ Cable routing
- ▶ Quick disconnect install

4.1 Overview of the install

This guide describes the harness cables and connectors for a generic dozer system, and applies to a new machine or when replacing components on a machine currently using grade control.

4.2 Connections, components, and cables

Part number	Items	Description
150603-02	1	EC520 long platform harness
150473-020	1	Cable extension, 2m
150473-040	1	Cable extension, 4m

4.3 Cable routing

One of the easiest ways to see how the system fits best on your machine is to drape the cables in possible locations, adding extension cables where needed and protecting any places where the cable will rub against edges.

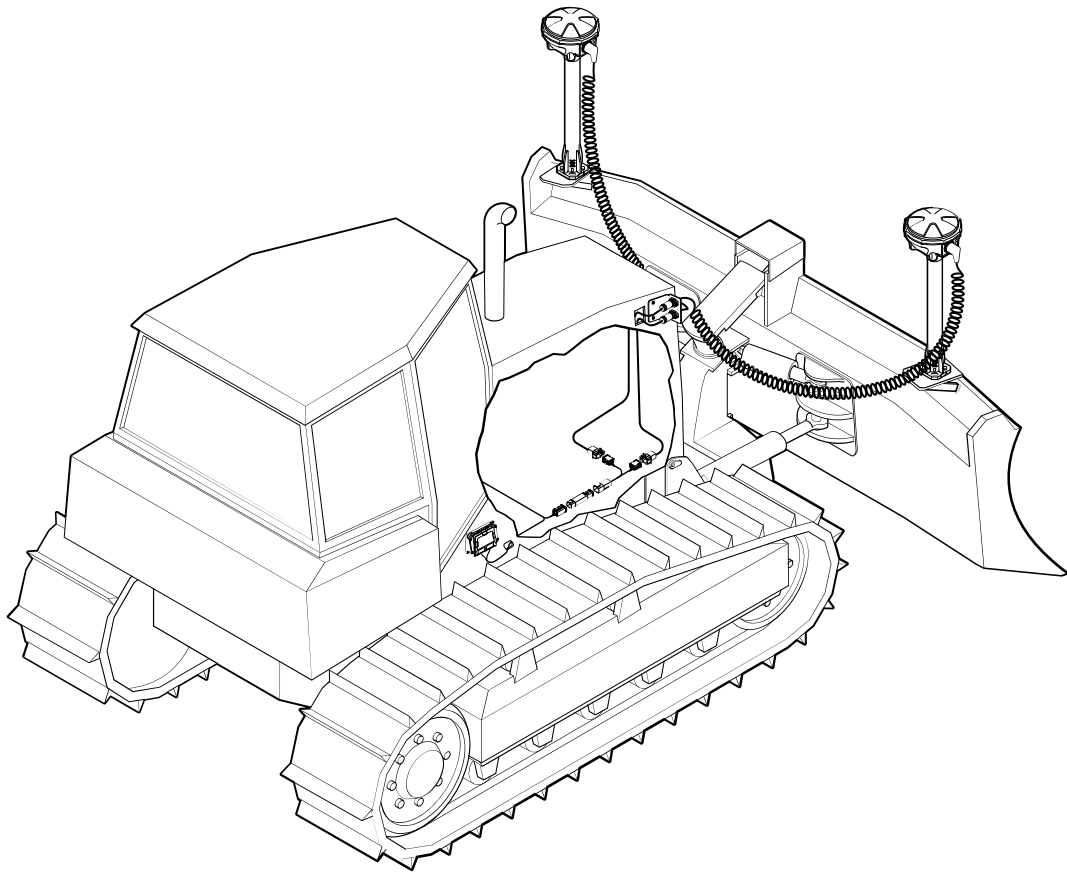
Using a combination of splitters (P/N 150858-01), inline terminators (P/N 150857-01), and the cables listed in [4.2 Connections, components, and cables](#), connect the system hardware to the EC520 long platform harness.

NOTE –Depending on the size of the machine, more or fewer extension cables (P/N 150416-0x0, where x indicates the length of the cable in meters) may be required. The Generic Dozer Base Kit provides one 4 m and one 2 m extension cables.

4.3.1 Routing the main harness

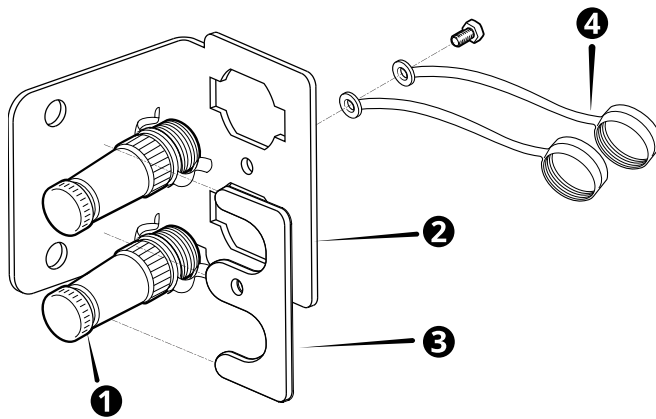
1. Starting from the front of the machine, slide the extension cable down the cable routing hole on the machine's hood, ensuring that the cable reaches the quick disconnect housing.
2. Route the cable across the front of the machine, above the coolant recovery tanks.
3. Continue routing the cable down the side of the machine, following the existing machine harness and use the CL-Clamps to secure the cable at regular intervals.
4. Route the cable through the engine compartment and into the right side access compartment.
5. Complete the connection to the EC520 platform harness, as shown in the connection diagram.

The following illustration shows a possible routing solution for a system with two masts on the blade.



4.4 Quick disconnect install

The following figure shows a dual quick disconnect bracket. The mounting bracket (❷) can be bolted or welded to the engine hood.



4.4.1 Quick disconnects connections, components, and cables

Part number	Items	Description
150624-02	4	Quick disconnects (❶)
185150-10	2	Quick disconnect bracket (❷)
0794-5930	2	Retainer clamp (❸)
	4	Dust caps (❹)

4.4.2 Install and connect quick disconnects

1. Mount the quick disconnect bracket to the engine hood.
2. Mount the quick disconnects into the bracket, aligning the spring clips with the openings at the top and bottom of the cutouts.
3. Insert the retainer clamp that holds the connectors in place.

NOTE –Insert the retainer clamp so that its thread insert faces the back of the bracket.

4. Secure the dust caps to the front of the bracket and clamp the connectors in place.
5. Connect the left/right splitter cable to the harness and route the connector to the quick disconnects.

Dozer mast install guide

In this guide:

- ▶ Overview of the install
- ▶ Components
- ▶ Install the mast mounts
- ▶ Installing the masts

5.1 Overview of the install

This guide describes how to install mast mounts on a dozer blade. When the mast mounts are installed, you can add:

- Fixed masts for a blade-mounted dual GNSS system.
- Electric masts for a blade-mounted dual laser receiver system.

Before you start to install the masts:

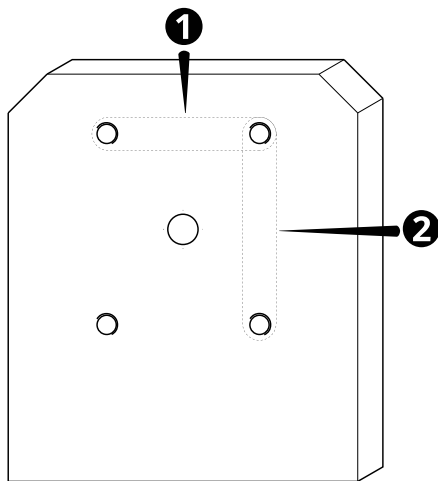
- Position the machine on a flat and level surface with the blade resting on blocks that are approximately the height of the track grouser and positioned so that the top of the cutting edge is level.
- Clean the threaded bolt holes in the blade. Failure to do so may prevent you from completely tightening the bolts to the mast mounts.
- Review this entire chapter. This will ensure that you understand the full installation workflow. It also lets you consolidate welding and fitting tasks, which may require external contractors, with other subsystem installations, such as the cab and blade installations.

5.2 Components

Part number	Required	Description
78003-02	3	Kit – Left and right mast mount
76702	1	Kit – Center mast mount

The mounting plates have five pre-drilled holes. Position the:

- 100 mm (3.94 inches) spaced holes (❶) parallel to the cutting edge.
- 125 mm (4.92 inches) spaced holes (❷) parallel to the machine centerline.



You may use the standoff in the kit to fabricate custom gussets to attach the mounting plate to the cutting edge. The gussets must be 12 mm – 19 mm (0.5 inches – 0.75 inches) thick.

5.3 Install the mast mounts

These instructions apply to left, center and right mast mounts. Specific locations are specified.

5.3.1 Position the left and right mast mounts

Position the left and right mast mount on top of the blade:

- as close as possible to each end of the cutting edge
- approximately an equal distance from the center of the blade

NOTE –The mast mount does not have to be flush with the top of the blade/cutting edge.

5.3.2 Position the center mounting block

Position the cutting edge in its normal operating position and then check that the mast mounts are:

- level in all directions (forward and back, left and right). Adjust if necessary.
- even with the back edge of the blade to prevent damage to the pitch link and mast mount.
- parallel to the cutting edge and in line with the side mast mounts.
- within +/- 200 mm of the true center of the blade. This is important for Autos performance.

NOTE –The mounting block does not have to be flush with the top of the blade/cutting edge.

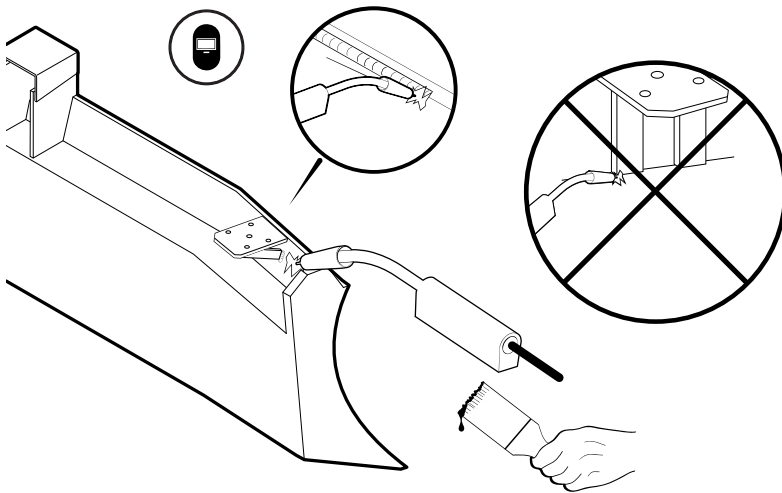
5.3.3 Weld the mast mounts

WARNING – Disconnect the negative battery terminal before commencing any welding and ensure that power is removed from the grade control system by disconnecting the power input cable or cables. Failure to follow this warning may result in electrical damage to system components.

TIP – Tack weld and check clearance to masts on the machine before welding completely. Raise the blade to full height and run through full tilt motion in all supported pitch positions (you may need to change the turnbuckle position) to ensure masts and receivers installed on top of the masts will not come into contact with parts of the machine body or lift rams, lights, etc.

If installing electric masts, ensure that the M4 shock mount has clearance to the blade lifting points before welding.

1. Tack weld the mounting plates to the cutting edge.
2. Tack weld the fabricated gussets to an even surface of the cutting edge.
3. Position the blade in the normal operating position with the cutting edge bolt line level from left to right. You may need to sight the blade on blocks to account for grouser penetration. Check that the mounting plates are level in all directions (forward and back, left and right). Adjust if necessary.
4. When everything is correct, weld the mount securely.
5. Once the welding is completed, paint the mounting bracket and weld area where machine paint was removed.



5.3.4 Install the optional GS520 sensor brackets - Inactive Body IMU dual mast systems

Refer to 98603-12 - *Install Manual for Dozer GS5XX Mast Mount Blade Sensor* that is included with the sensor bracket kit.

5.4 Installing the masts

For instructions on installing masts for a blade-mounted system, see the [Chapter 6, Dozer GNSS install guide](#).

Tips for connecting mast cables, connect:

- Left mast cables to the left quick disconnect on the front of the machine.
- Right mast cables to the right quick disconnect on the front of the machine.
- Center mast cables to the left quick disconnect on the front of the machine.
- GS5xx sensor cables to any quick disconnect on the front of the machine.

Dozer GNSS install guide

In this guide:

- ▶ Overview of the install
- ▶ GNSS connections, components, and cables
- ▶ GNSS mast and receiver install
- ▶ Validate the GNSS system install

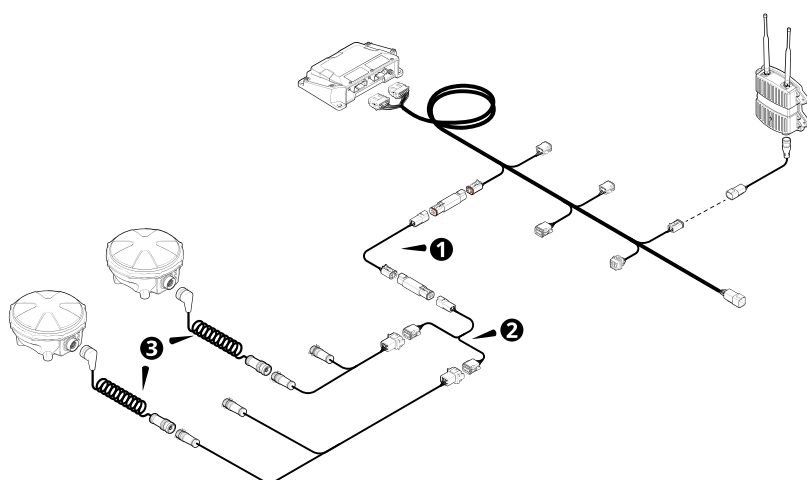
6.1 Overview of the install

This guide supports the addition of the hardware components of a GNSS positioning subsystem to a base Earthworks dozer system.

Before starting a GNSS subsystem install:

- Complete the installation of the cab components. In particular, if you are installing the optional radio then you need to install the SNM941 Gateway stacked radio bracket first.
- Review this entire guide. This will ensure that you understand the full installation workflow. It also lets you consolidate welding and fitting tasks, which may require external contractors, with other subsystem installations, such as the cab and front linkage installations.

6.2 GNSS connections, components, and cables



Item	Part number	Description
❶	150416-0x0	Extender cable
❷	150750-01	GNSS splitter cable
❸	150413-0xx	If required, GNSS Smart Antenna cables.

6.2.1 Components and cables from base kit (P/N 160005-500)

Part number	Provided	Description
150416-0x0	2	Extender cable (❶)
150606-01	2	Splitter
150857-01	4	Terminator
150750-01	1	GNSS splitter cable (❷)

6.2.2 Additional MS995 Smart Antenna components and cables

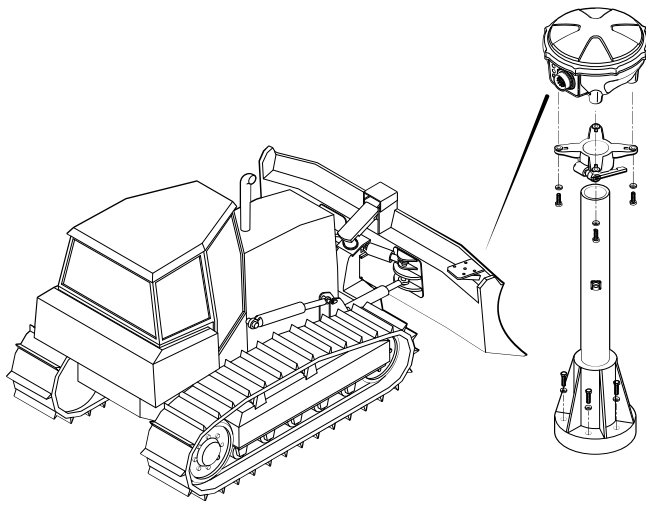
Part number	Required	Description
95106-01	2	MS995 GNSS Smart Antenna with mounting bracket
150413-0xx	2	Smart Antenna cables (❸)
81069-10	2	Steel mast

6.3 GNSS mast and receiver install

Components associated with the GNSS receiver masts installation include:

- MS99x GNSS receiver
- Mast, P/N 81069-10
- Mast mounting clamp assembly P/N 52070-00/10

CAUTION – Mount GNSS receivers only on the mast included in the kit and only as supplied. Using other masts or altering the supplied masts damages the receivers.



To install:

1. Attach the clamp assembly to the bottom of the receiver and torque the quick disconnect bracket bolts to $41 \text{ Nm} \pm 8\%$.
2. Slide the receiver with the clamp assembly onto the top of the mast and tighten the clamp.
3. Use the strain relief bracket to secure the receiver cable.
4. Plug the coil cable into the quick disconnect.

6.4 Validate the GNSS system install

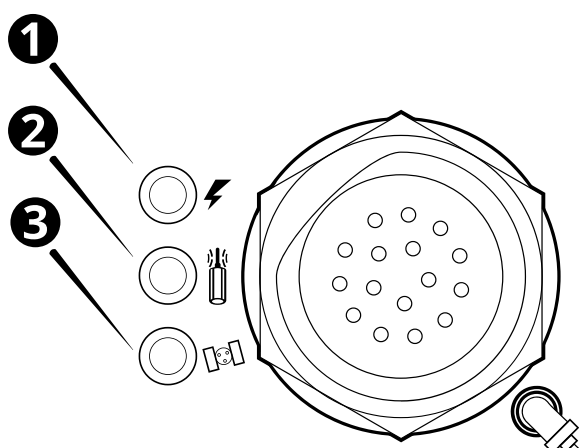
To validate the GNSS system installation:

- [6.4.1 Check GNSS receiver status indicators](#)
- [6.4.2 Check radio power LED](#)
- [6.4.3 Check device connections](#)

6.4.1 Check GNSS receiver status indicators

The MS9xx GNSS receivers have three LEDs next to the harness connector. How these LEDs behave indicates the receiver status, as shown in the following figure.

In a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be solid.

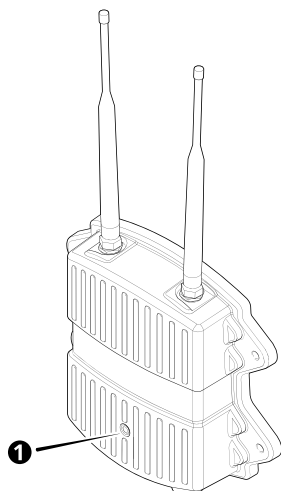


LED	Left	Right
❶ Power	Off – No power On – Power *	Off – No power On – Power *
❷ Data link	Off – No corrections 1 Hz – Corrections [†]	Off – No corrections On – Corrections received from the left receiver [†]
❸ Satellite	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *

* required

[†] in a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be on (solid)

6.4.2 Check radio power LED



The radio is connected to power if the LED is permanently on or flashing.

6.4.3 Check device connections

In the Web Interface, navigate to Monitor > Onboard Devices, and check that the newly installed devices are shown as connected.

Supported third party tablet installation

In this chapter:

- ▶ Overview of third-party tablet installation
- ▶ Third-party tablet installation

7.1 Overview of third-party tablet installation

7.1.1 Purpose of this guide

The purpose of this guide is to enable the use of supported third-party tablets with the Earthworks system.

7.1.2 Supported third-party tablets

To run this release of the product on a third-party tablet, use a tablet that meets the requirements for Earthworks. Refer to the Earthworks release notes for more information.

7.1.3 Prerequisites

Before starting a third-party tablet installation:

- Make sure that the EC520 controller has the required OS and firmware installed, and that the controller's Wi-Fi access point is enabled. For more information, refer to *Earthworks Configure and Connect Guide*.
- It is strongly recommended that you review this entire guide. This will ensure you understand the full installation workflow.

7.1.4 Equipment list

NOTE –For users who supply their own display, the kit still includes a TD5x0 display harness, which must be used for connecting to the AA510.

- A PC
- Optionally, a cable to connect the tablet to a cigarette lighter for power

7.2 Third-party tablet installation

7.2.1 Download the app to the tablet

1. Turn on the tablet.
2. For Android 6 or 7 devices, navigate to Settings > Security > Unknown Source, and allow installation of apps from unknown sources.
3. Navigate to Settings > WLAN and connect to the EC520 controller:
 - The name of the access point is the SSID set during the configuration of the controller.
 - The password of the access point is the password set during the configuration of the controller.

NOTE – Some implementations of the Android operating system do not automatically reconnect to Wi-Fi access points that are not connected to the Internet. If a Network Error message displays after you turn the tablet off, and then turn it on again, check that Wi-Fi is enabled on the tablet, and that the controller is connected.

4. In a web browser, browse to: 192.168.168.1. The Web Interface log-in screen displays.

TIP – Earthworks supports the Google Chrome browser to run the Web Interface.

5. Log in to the Web Interface. The Web Interface Home screen displays.
6. Navigate to Advanced > Install Operator App. The Install Operator App screen displays.
7. Select Install Direct. The Earthworks firmware download begins.
8. On your device, locate the downloaded *launcher.apk* file, and select it.
9. Follow the instructions on the screen. For Android 8 and above, if the install is blocked by Android, set Allow from this source to ON.
10. If prompted, allow the plugin to access photos, media, and files on your device.

7.2.2 Locate and attach the tablet

In the cab:

1. Decide on how to attach the supported third-party tablet in the cab. We do not suggest locations or methods of attachment.
2. Optionally, attach the cable to connect the display to a cigarette lighter for power.

7.2.3 Validate the install

NOTE – To view the log-in screen, make sure at least one operator is configured, and that maintenance lockout is not active. For more information, refer to the *Configuration Guide*.

7 Supported third party tablet installation

1. Turn on the tablet.
2. Locate and tap the app icon. If one of the following displays, then the tablet is connected to the EC520 controller:
 - the log-in screen
 - the *No operator configured for this machine* error message
 - the *Maintenance Lockout Active* error message

It is recommended that you install Android updates when available to keep the system up to date.

Torque charts

8.1 Imperial

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

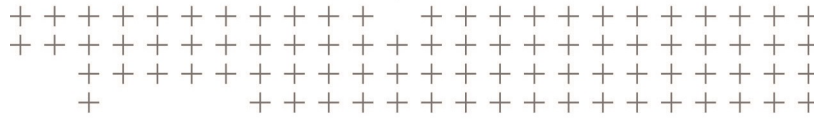
Nominal size (inches)											
	Grade	Torque	1/4	5/16	3/8	7/16	1/2	9/16	5/8	3/4	1"
	SAE 2	FT-LB	6	11	19	30	45	66	93		300
										150	
	SAE 5	FT-LB	9	18	31	50	75	110	150	250	580
		Nm	12	24	42	68	102	149	203	339	786
	SAE 8	FT-LB	13	29	46	75	115	165	225	370	890
		Nm	18	39	62	102	156	224	305	502	1207
	SOCKET HEAD	FT-LB	14	30	50	80	120	175	240	390	960
		Nm	19	41	68	108	163	237	325	529	1302

8.2 Metric

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

Nominal size (mm)										
	Grade	Torque	6	8	10	12	14	16	18	20
	4.6, 4.8	FT-LB	3	8	16	27	43	60	90	120
		Nm	4	11	22	37	58	81	122	163
	8.8	FT-LB	7	17	33	59	100	130	180	280
		Nm	10	23	45	80	136	176	244	380

Nominal size (mm)										
	Grade	Torque	6	8	10	12	14	16	18	20
	10.9	FT-LB	10	25	48	85	130	200	280	400
		Nm	14	34	65	115	176	271	380	542
	12.9	FT-LB	12	29	57	100	150	230	–	–
		Nm	16	39	77	136	203	312	–	–



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